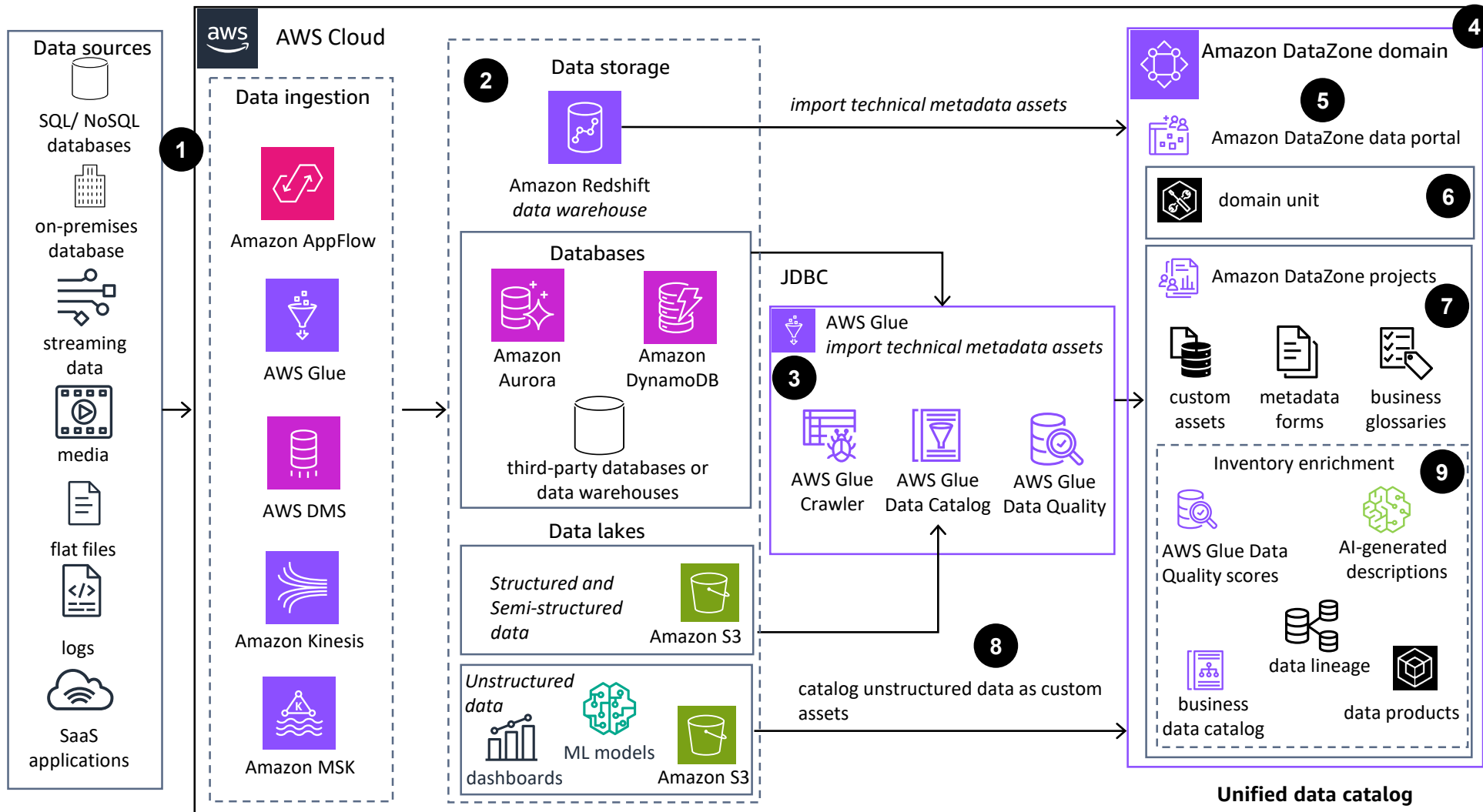


Building a unified catalog with Amazon DataZone

The unified data catalog acts as a central repository for all your organization's data assets. It goes simply beyond listing data sources; it empowers users with rich context and insights about the data they encounter. This architecture shows how Amazon DataZone breaks down the data silos within the organization by bringing data from various sources together and fostering improved search functionality and trust in data.



- 1 Data is extracted using Amazon AppFlow, AWS Glue, Amazon Kinesis, Amazon Managed Streaming for Apache Kafka (Amazon MSK) or AWS Database Migration Service (AWS DMS).
- 2 Based on the end user requirements, the data is stored in Amazon Simple Storage Service (Amazon S3), Amazon Redshift, or purpose-built databases like Amazon Aurora or Amazon DynamoDB. Data lakes are designed to store raw structured, semi-structured and unstructured data at low cost.
- 3 Structured and semi-structured data from Amazon S3 is further crawled using AWS Glue Crawler, which writes the metadata to AWS Glue Data Catalog. You can perform data quality checks on these catalog tables at rest using AWS Glue Data Quality. Crawler uses Java Database Connectivity (JDBC) data sources to crawl and catalog the data.
- 4 Amazon DataZone domain acts as the central catalog repository hub. AWS Glue Data Catalog data source can be used to onboard the existing data assets from data lakes and databases to Amazon DataZone. With Amazon Redshift, the technical metadata of the database tables and views from the existing data warehouse is automatically extracted to Amazon DataZone.
- 5 Use Amazon DataZone data portal to discover, catalog, share and govern data in a self-serve fashion.
- 6 Amazon DataZone entities can be organized under different levels of hierarchy using domain units.
- 7 Amazon DataZone data projects provide a collaborative space where members within an organization can onboard data assets from their respective business units. Business glossaries is a dictionary of business terms that may be associated with data assets. Metadata forms defines additional information to augment business context of the asset metadata. Custom assets gives flexibility to expand the catalog beyond pre-defined systems assets (for e.g. ML models, images, videos).
- 8 Unstructured data assets stored in Amazon S3 can be catalogued and published to Amazon DataZone by tagging them with relevant custom asset type.
- 9 Every data asset onboarded to Amazon DataZone will be part of Inventory. The data asset is further enriched with business catalogs, data lineage, data quality to aid discoverability of the asset.



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AWS Reference Architecture